

B.E. – Computer Science and Engineering

Laboratory Record

U21CS504

Web Technologies Laboratory

Regulation 2021

Nov / Dec 2026

Department of Computer Science and Engineering
KPR Institute of Engineering and Technology



Name : _____

Roll Number : _____

Subject Code & Title : _____

Department : _____

Year & Semester : _____

This is the certified record of work done by.....

Register Number

Faculty In- Charge

Head of the Department

Place:

Date:

He/ She has submitted the record for the End Semester Practical Examination held on

Internal Examiner

External Examiner

List of Experiments

Expr. No.	Name of the Experiments	Page No.	Marks (20)	Faculty Sign
1.	Static web page design using HTML and CSS			
2.	BMI Calculator using JavaScript			
3.	User registration form validation using JavaScript			
4.	Implementing web applications with API calls using Java Script			
5.	Event registration form using PHP and MySQL			
6.	Web application for food delivery system using PHP and MySQL			
7.	Web application for leave management using Servlet and MySQL			
8.	Web application for online shopping using Servlet and MySQL			
9.	Mini Project:			

Rubrics for Laboratory Experiments

Criteria & Points	Strongly Acceptable	Acceptable	Considerable	Below Expectations	Unacceptable
Content	4	3	2	1	0
Understanding the Problem Statement	The procedure is explained and well designed the problem with appropriate analysis	Some programs are written accurately	Major errors in program writing	Programs are not written/wrong program written	Programs are not written/wrong program written
Completion of target in Lab	100% target has been completed	80% target has been completed	70% target has been completed	50% target has been completed	25% target has been completed
User Interface Design	Well designed problem with appropriate analysis	Designed the problem with analysis	Missing some experimental procedure with partial analysis	Missing major experimental details and analysis	Could not be able to explain the program
Development	Implement Optimal solution with appropriate results	80% of implementation has been done	65% of implementation has been done. Few procedures were explained	No proper explanation on implementation	Could not be able to explain the program
Innovation	Most of the innovations are applied clearly	Some of the innovations are applied undoubtedly	Few innovations are applied clearly	innovations applied	None of the innovations used

AIM:

To design and develop a basic multi-page **book store website** using HTML and CSS. The website includes Home, About, Books, and Contact pages with a common navigation menu, images, tables, forms, and CSS styling.

PROCEDURE:

1. Create a project folder named **BookNestWebsite**.
2. Create four HTML files:
 1. **index.html**
 2. **about.html**
 3. **services.html**
 4. **contact.html**
3. Create a CSS file named **style.css**.
4. Design the basic HTML structure for each page using:
 1. `<!DOCTYPE html>`
 2. `<html>`
 3. `<head>`
 4. `<title>`
 5. `<link>`
 6. `<body>`
5. Create a common navigation bar using the `<nav>` and `<a>` tags to connect all pages.
6. Design the **Home** page with:
 1. Book Store Name
 2. Logo
 3. Welcome Message
 4. Banner Image
 5. Featured Books
7. Design the **About** page with:
 1. Store Profile
 2. Vision
 3. Mission
 4. Objectives
 5. Staff Information
8. Design the **Books** page with:
 1. Book Categories

2. Book Details
3. Book Images
4. Price Table

9. Design the **Contact** page with:

1. Address
2. Phone Number
3. Email
4. Contact Form

10. Apply CSS styling to improve the appearance by adding:

- Background colors
- Fonts
- Borders
- Padding
- Margins
- Navigation styling
- Hover effects
- Table styling
- Form styling

1. Save all files.
2. Open **index.html** using a web browser or Live Server.
3. Verify that all hyperlinks, images, and styles are working correctly.

HTML ELEMENTS USED:

- `<!DOCTYPE html>` – Declares the HTML5 document.
- `<html>` – Root element of the webpage.
- `<head>` – Contains metadata.
- `<title>` – Displays the page title.
- `<link>` – Connects the CSS file.
- `<body>` – Contains webpage content.
- `<nav>` – Creates the navigation bar.
- `<a>` – Creates hyperlinks.
- `<h1>`, `<h2>` – Headings.
- `<p>` – Paragraph.
- `<img>` – Displays images.
- `<ul>` and `<li>` – Creates unordered lists.
- `<table>` – Displays book price details.
- `<tr>`, `<th>`, `<td>` – Creates rows and columns in a table.
- `<form>` – Creates the contact form.
- `<label>` – Defines labels for form elements.
- `<input>` – Accepts user input.
- `<textarea>` – Accepts multi-line text.
- `<button>` – Creates clickable buttons.
- `<footer>` – Defines a footer for the document.
- `<main>` – Defines the main content area.

CSS PROPERTIES USED:

- font-family
- margin
- margin-top
- margin-bottom
- margin-left
- line-height
- display
- flex-direction
- min-height
- background-color
- padding
- text-align
- color
- text-decoration
- font-weight
- flex
- max-width
- width
- border
- border-radius
- cursor
- font-size

IMPLEMENTATION:

Index.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>BookNest Book Store</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

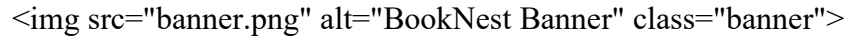
```
  <nav>
```

```
    <a href="index.html">Home</a>
```

[About](about.html)

[Services](services.html)

[Contact](contact.html)



BookNest Book Store



Welcome

Welcome to BookNest Book Store, your one-stop destination for book lovers.

We provide a wide collection of novels, academic books, children's books, biographies, and self-improvement books at affordable prices.

Why Choose Us?

- Large collection of books
- Affordable prices
- Latest bestsellers available
- Fast delivery service
- Friendly customer support

© 2026 BookNest Book Store. All Rights Reserved.

</footer>

</body>

</html>

Services.html

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Services - BookNest</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<nav>

Home

About

Services

Contact

</nav>

<main>

<h1>Our Services</h1>

<p class="page-intro">Enjoy a smooth and friendly book-shopping experience with services designed for every reader.</p>

<div class="cards">

<div class="card">

<h3>Book Collections</h3>

<p>Explore fiction, nonfiction, children's books, academic titles, and bestsellers.</p>

</div>

<div class="card">

<h3>Quick Delivery</h3>

<p>We offer fast and reliable delivery so your next favorite book arrives without delay.</p>

```
</div>
<div class="card">
  <h3>Personal Recommendations</h3>
  <p>Our team helps you discover books that match your interests and reading goals.</p>
</div>
</div>
</main>

<footer>
  <p>&copy; 2026 BookNest Book Store. All Rights Reserved.</p>
</footer>
</body>
</html>
```

About.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About - BookNest</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <nav>
    <a href="index.html">Home</a>
    <a href="about.html">About</a>
    <a href="services.html">Services</a>
    <a href="contact.html">Contact</a>
  </nav>

  <main>
    <h1>About BookNest</h1>
    <p class="page-intro">We are passionate about helping readers discover meaningful books in every genre.</p>
```

```
<div class="cards">
  <div class="card">
    <h3>Our Story</h3>
    <p>BookNest began as a small reading corner and grew into a vibrant bookstore with a warm,
welcoming spirit.</p>
  </div>
  <div class="card">
    <h3>Our Mission</h3>
    <p>We aim to make reading enjoyable, affordable, and accessible for every learner and book
lover.</p>
  </div>
  <div class="card">
    <h3>What We Offer</h3>
    <p>From novels to study guides, we bring together timeless classics and fresh releases for all
tastes.</p>
  </div>
</div>
</main>

<footer>
  <p>&copy; 2026 BookNest Book Store. All Rights Reserved.</p>
</footer>
</body>
</html>
```

Contact.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Contact - BookNest</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <nav>
```

Home

About

Services

Contact

</nav>

<main>

<h1>Contact Us</h1>

<p class="page-intro">We would love to hear from you. Reach out for book suggestions, orders, or feedback.</p>

<div class="contact-card">

<h3>Get in Touch</h3>

<p>Email: support@booknest.com</p>

<p>Phone: +1 234 567 890</p>

<p>Address: 12 Reading Street, Booktown</p>

<form>

<label for="name">Name</label>

<input type="text" id="name" placeholder="Your name">

<label for="email">Email</label>

<input type="email" id="email" placeholder="Your email">

<label for="message">Message</label>

<textarea id="message" rows="4" placeholder="Write your message"></textarea>

<button type="submit">Send Message</button>

</form>

</div>

</main>

<footer>

<p>© 2026 BookNest Book Store. All Rights Reserved.</p>

</footer>

```
</body>
```

```
</html>
```

style.css:

```
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}
```

```
body {  
  font-family: Arial, sans-serif;  
  background: #f7f4ef;  
  display: flex;  
  flex-direction: column;  
  min-height: 100vh;  
  line-height: 1.6;  
}
```

```
/* Navigation */
```

```
nav {  
  background: #5d4037;  
  padding: 15px;  
  text-align: center;  
}
```

```
nav a {  
  color: white;  
  text-decoration: none;  
  margin: 0 20px;  
  font-weight: bold;  
  font-size: 18px;  
}
```

```
nav a:hover {  
    color: #ffd54f;  
}
```

```
/* Banner */
```

```
.banner {  
    width: 90%;  
    max-width: 1000px;  
    height: 260px;  
    object-fit: cover;  
    display: block;  
    margin: 20px auto;  
    border-radius: 10px;  
    border: 3px solid #6d4c41;  
}
```

```
/* Main Section */
```

```
main {  
    width: 85%;  
    max-width: 900px;  
    margin: auto;  
    background: white;  
    padding: 30px;  
    border-radius: 10px;  
    box-shadow: 0 0 10px rgba(0, 0, 0, 0.15);  
}
```

```
/* Logo */
```

```
.logo {  
    width: 180px;  
    height: 180px;
```

```
    object-fit: contain;
    display: block;
    margin: 20px auto;
}
```

```
/* Headings */
```

```
h1 {
    text-align: center;
    color: #5d4037;
    margin-bottom: 15px;
}
```

```
h2 {
    color: #8d6e63;
    margin-top: 20px;
    margin-bottom: 10px;
}
```

```
/* Paragraph */
```

```
p {
    text-align: justify;
    margin-bottom: 15px;
}
```

```
/* List */
```

```
ul {
    margin-left: 25px;
}
```

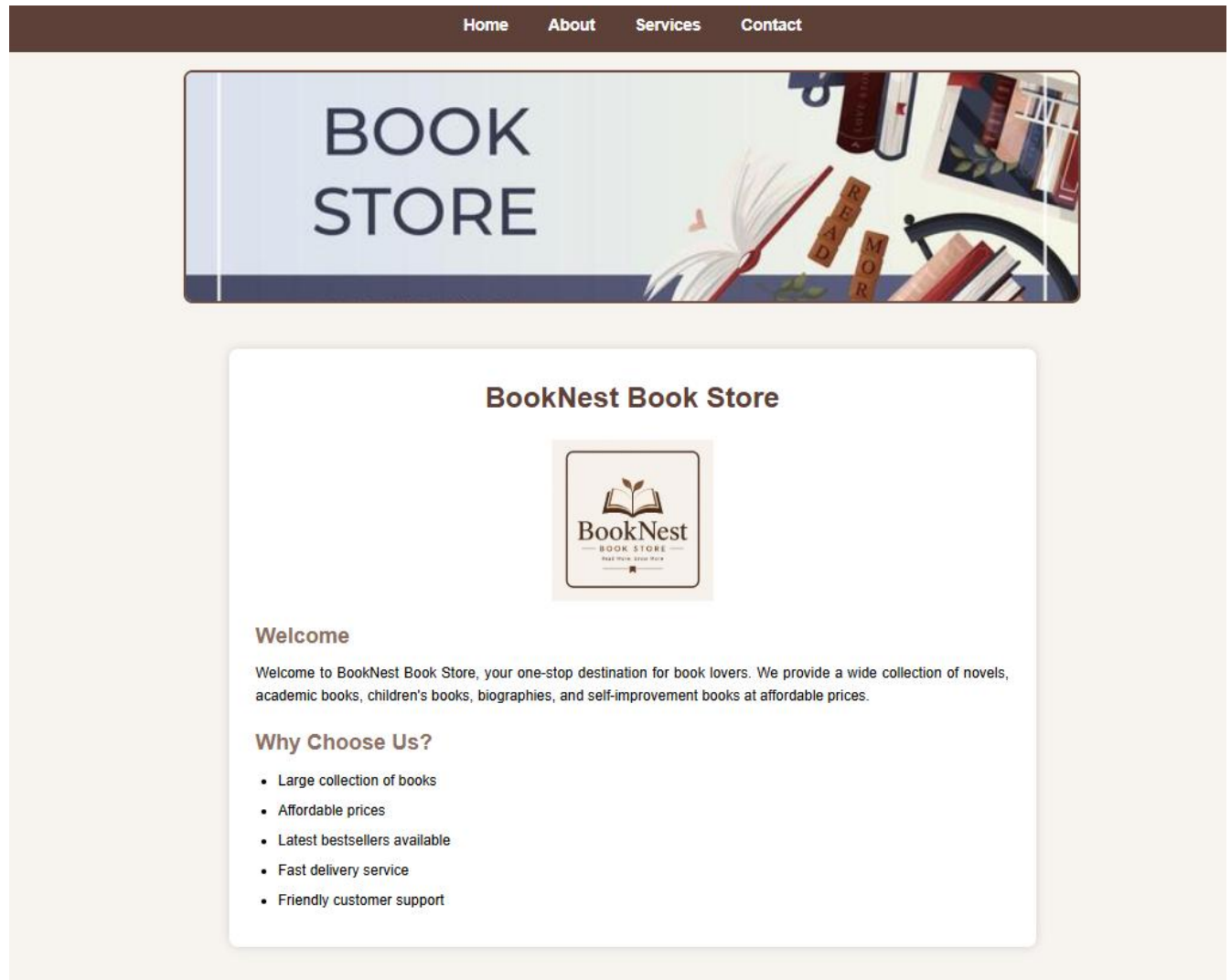
```
li {
    margin-bottom: 8px;
}
```

```
/* Footer */
```

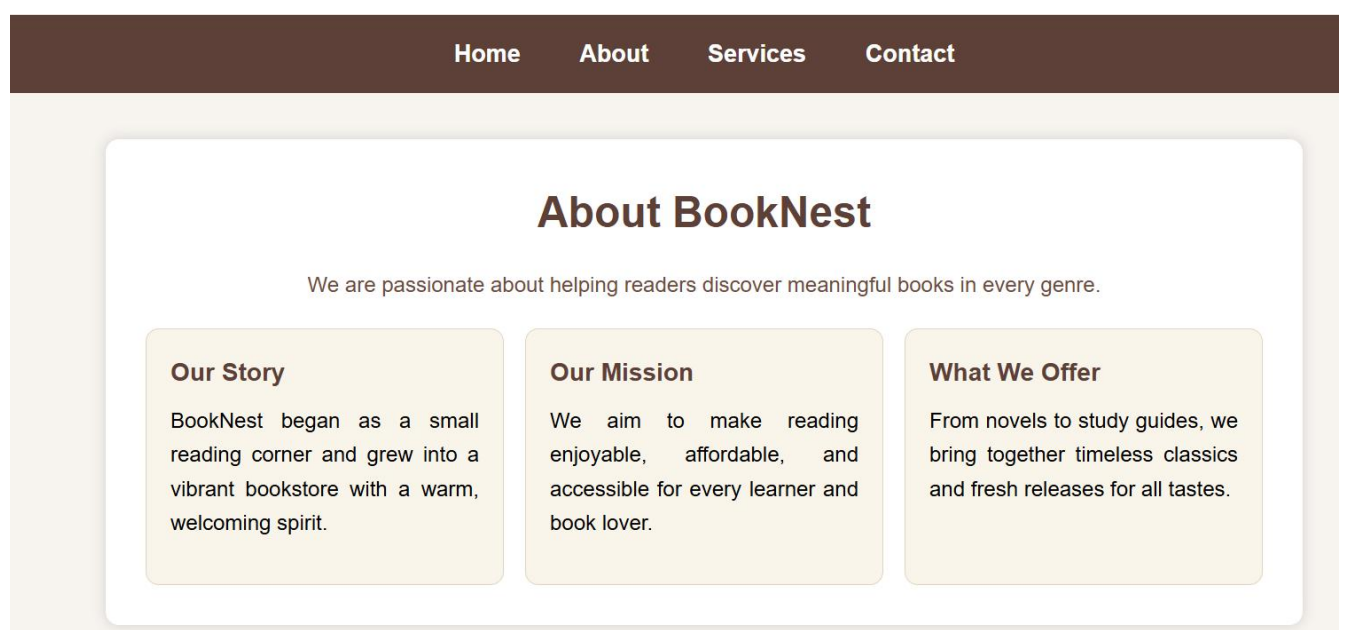
```
footer {  
  background: #5d4037;  
  color: white;  
  text-align: center;  
  padding: 15px;  
  margin-top: 30px;  
}
```


OUTPUT:

1. Home Page



2. About Page



Service page

[Home](#) [About](#) [Services](#) [Contact](#)

Our Services

Enjoy a smooth and friendly book-shopping experience with services designed for every reader.

Book Collections

Explore fiction, nonfiction, children's books, academic titles, and bestsellers.

Quick Delivery

We offer fast and reliable delivery so your next favorite book arrives without delay.

Personal Recommendations

Our team helps you discover books that match your interests and reading goals.

Contact page

Contact Us

We would love to hear from you. Reach out for book suggestions, orders, or feedback.

Get in Touch

Email: support@booknest.com

Phone: +1 234 567 890

Address: 12 Reading Street, Booktown

Name

Email

Message

[Send Message](#)

RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Understanding (R1)	4	
Timeline (R2)	4	
Design (R3)	4	
Development (R4)	4	
Innovation (R5)	4	
Total Marks	20	

RESULT:

A basic multi-page company website was successfully designed and developed using HTML and CSS. The website contains Home, About, Products/Services, and Contact pages connected through a common navigation menu. CSS styling was applied to improve the appearance, and all hyperlinks, images, tables, and forms function correctly.

Ex.No: 02

Date: 15//07/2026

BMI CALCULATOR USING JAVASCRIPT

AIM:

To design and develop an interactive **HealthTrack BMI Calculator** using HTML, CSS, and JavaScript that accepts the user's weight and height, validates the input values, calculates the Body Mass Index (BMI), displays the BMI category, and provides a health recommendation without refreshing the webpage.

PROCEDURE:

1. Create a project folder named **HealthTrackBMI**.
2. Create the following files:
 1. **index.html**
 2. **style.css**
 3. **script.js**
3. Design the webpage structure in **index.html** using HTML elements such as headings, labels, input fields, buttons, and containers.
4. Link the external CSS file using the <link> tag.
5. Link the JavaScript file using the <script> tag.
6. Create input fields to receive:
 1. Weight (kg)
 2. Height (cm)
7. Add **Calculate BMI** and **Clear** buttons.
8. Design the webpage using CSS by applying:
 1. Background color
 2. Fonts
 3. Padding
 4. Margins
 5. Borders
 6. Border Radius
 7. Button Styling
 8. Hover Effects
 9. Box Shadow
9. Write JavaScript code to:
 1. Read user inputs.
 2. Validate the entered values.
 3. Calculate BMI using the formula:
 $BMI = Weight / (Height \times Height)$
 4. Display the BMI value rounded to two decimal places.

5. Determine the BMI category.
6. Display an appropriate health suggestion.
10. Display all results dynamically without reloading the webpage.
11. Save all files.
12. Open **index.html** using a web browser or Live Server.
13. Test the application using different input values and verify the output.

HTML ELEMENTS USED:

- `<!DOCTYPE html>` – Declares the HTML5 document.
- `<html>` – Root element of the webpage.
- `<head>` – Contains metadata.
- `<meta>` – Defines character encoding and viewport settings.
- `<title>` – Displays the page title.
- `<link>` – Connects the CSS file.
- `<body>` – Contains webpage content.
- `<div>` – Groups webpage elements.
- `<h1>` – Displays the main heading.
- `<label>` – Defines labels for input fields.
- `<input>` – Accepts user input.
- `<button>` – Performs actions.
- `<p>` – Displays result messages.
- `<script>` – Links the JavaScript file.

CSS PROPERTIES USED:

- `font-family`
- `background-color`
- `color`
- `width`
- `padding`
- `margin`
- `border`
- `border-radius`
- `box-shadow`
- `display`
- `text-align`
- `cursor`
- `font-size`
- `font-weight`
- `transition`
- `box-sizing`

JAVASCRIPT FEATURES & METHODS USED:

- function
- let
- const
- document.getElementById()
- .value
- .innerHTML
- parseFloat()
- isNaN()
- if...else
- return
- .toFixed()
- Arithmetic Operators (+, -, *, /)
- Comparison Operators (<, >, <=)

IMPLEMENTATION:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HealthTrack BMI Calculator</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
  <h1>HealthTrack BMI Calculator</h1>

  <label for="weight">Weight (kg)</label>
  <input type="number" id="weight" placeholder="Enter your weight">

  <label for="height">Height (cm)</label>
  <input type="number" id="height" placeholder="Enter your height">

  <button onclick="calculateBMI()">Calculate BMI</button>
  <button class="clear" onclick="clearFields()">Clear</button>

  <div class="result">
```

```
<h2>Your Result</h2>
<p id="bmiValue">BMI: --</p>
<p id="status">Category: --</p>
<p id="advice"></p>
</div>
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Style.css:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: Arial, sans-serif;
}

body {
  background: #e8f5e9;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}

.container {
  width: 380px;
  background: white;
  padding: 30px;
  border-radius: 12px;
  box-shadow: 0 0 15px rgba(0, 0, 0, 0.2);
}
```

```
h1 {  
  text-align: center;  
  color: #2e7d32;  
  margin-bottom: 20px;  
}
```

```
label {  
  display: block;  
  margin-top: 15px;  
  margin-bottom: 5px;  
  font-weight: bold;  
}
```

```
input {  
  width: 100%;  
  padding: 10px;  
  border: 1px solid #999;  
  border-radius: 6px;  
  font-size: 16px;  
}
```

```
button {  
  width: 100%;  
  padding: 12px;  
  margin-top: 15px;  
  border: none;  
  border-radius: 6px;  
  background: #2e7d32;  
  color: white;  
  font-size: 16px;  
  cursor: pointer;  
}
```

```
button:hover {
```



```
    background: #1b5e20;
}
```

```
.clear {
    background: #c62828;
}
```

```
.clear:hover {
    background: #8e0000;
}
```

```
.result {
    margin-top: 25px;
    text-align: center;
    background: #f5f5f5;
    padding: 15px;
    border-radius: 8px;
}
```

```
.result h2 {
    margin-bottom: 10px;
    color: #333;
}
```

```
.result p {
    margin: 8px 0;
    font-size: 18px;
}
```

script.js:

```
function calculateBMI() {

    let weight = parseFloat(document.getElementById("weight").value);
    let height = parseFloat(document.getElementById("height").value);
```

```
if (isNaN(weight) || isNaN(height) || weight <= 0 || height <= 0) {  
    alert("Please enter valid weight and height.");  
    return;  
}
```

```
let heightMeter = height / 100;
```

```
let bmi = weight / (heightMeter * heightMeter);
```

```
let category = "";
```

```
let advice = "";
```

```
if (bmi < 18.5) {  
    category = "Underweight";  
    advice = "You should eat a balanced diet and gain healthy weight.";  
}  
else if (bmi < 25) {  
    category = "Normal";  
    advice = "Great! Maintain your healthy lifestyle.";  
}  
else if (bmi < 30) {  
    category = "Overweight";  
    advice = "Exercise regularly and follow a healthy diet.";  
}  
else {  
    category = "Obese";  
    advice = "Consult a healthcare professional for proper guidance.";  
}
```

```
document.getElementById("bmiValue").innerHTML = "BMI: " + bmi.toFixed(2);  
document.getElementById("status").innerHTML = "Category: " + category;  
document.getElementById("advice").innerHTML = advice;  
}
```

```
function clearFields() {
```

```
document.getElementById("weight").value = "";
```

```
document.getElementById("height").value = "";
```

```
document.getElementById("bmiValue").innerHTML = "BMI: --";
```

```
document.getElementById("status").innerHTML = "Category: --";
```

```
document.getElementById("advice").innerHTML = "";
```

```
}
```

OUTPUT:

VALID CASE:

HealthTrack BMI Calculator

Weight (kg)

56

Height (cm)

160

Calculate BMI

Clear

Your Result

BMI: 21.87

Category: Normal

Great! Maintain your healthy lifestyle.

**INVALID
CASE:**

127.0.0.1:5500 says

Please enter valid weight and height.

OK

-100

Height (cm)

182

Calculate BMI

Clear

Your Result

BMI: 30.19

Category: Obese

Consult a healthcare professional
for proper guidance.

RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Understanding (R1)	4	
Timeline (R2)	4	
Design (R3)	4	
Development (R4)	4	
Innovation (R5)	4	
Total Marks	20	

RESULT:

The BMI Calculator application was successfully designed, developed, and tested using HTML, CSS, and JavaScript. The system accurately validates user inputs, computes the BMI rounded to two decimal places, assigns the correct health category with a suitable advice message, and dynamically updates the DOM without reloading the page.